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FERTILITY TRENDS IN SINGAPORE

KULDIP SINGH, OSBORN VIEGAS AND S. S. RATNAM

Department of Obstetrics and Gynaecology, National University Hospital, Singapore

Summary. A major cause for the disparity in maternal and infant mortality and morbidity between the developing world and industrialized nations is the uncontrolled population growth seen in the former, largely brought about by failure of authorities to realize the importance of fertility regulation. Some governments and international agencies have introduced family planning programmes which have had a striking effect on the health of mothers and children. This study traces demographic changes in Singapore in the context of legislation disincentives and incentives.

Introduction

In general, National Family Planning programmes even in countries with explicit demographic goals rely largely on voluntary acceptance of contraception. But voluntary family planning programmes, even when highly successful, may often fail to achieve demographic goals (David, 1967). In these situations, measures may be recommended that go 'beyond family planning' in an attempt to influence the number of children couples choose to have (Berelson, 1969). Singapore is an example of a country that has gone well beyond voluntary family planning to reach its population objectives. Since 1966, the Singapore National Family Planning and Population programme has had very clear demographic objectives, i.e. to reach replacement reproduction by 1990 and achieve zero population growth by 2030 (Wan & Loh, 1973). To achieve these goals, the government of Singapore used legislation to discourage births above given birth orders: initially fourth births and later third births. It also introduced a series of incentives and disincentives listed in Table 1. Although these disincentives are economically appreciable, it must be stressed that the government does not otherwise penalize those not restricting family size to the stipulated two children.

In addition, the abortion laws were progressively liberalized to a point where termination of pregnancy is now available 'on demand' up to the 24th week of pregnancy, as follows.

Before 1967. Legal abortion was restricted only to those cases in whom maternal life was endangered.

1967. Medical committee of the National Family Planning and Population Board was set up: abortion was extended to: (1) cases with congenital fetal malformations;

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Table 1. Policies to reduce fertility

Disincentives	Incentives
1. Increasing accouchement charges for increasing birth orders	1. Waiver of delivery charges
2. Low school admission priority for families with three or more children	2. Paid medical leave for mothers undergoing sterilization after birth of third or subsequent children
3. No paid maternity leave for birth of third and subsequent children	3. Monetary stipends in some cases where mother is sterilized after birth of first or second child
4. Tax relief for first three children only	4. Continuation of employment for non-Singaporeans only if both partners undergo sterilization after birth of second child.
5. Low housing subsidy allocation priority for families with three or more children	

(2) cases where the woman was a victim of a sex crime or there had been intercourse with an insane or feeble-minded person.

1968. First Abortion Bill: abortion was further extended to cases deemed unsuitable for continuing pregnancy for family, social and economic reasons. This facility was only available to those women resident in Singapore for more than 4 months prior to abortion.

1974. New Abortion Act: this made it possible for abortions to be performed up to 24 weeks of pregnancy, at the written request of the woman, by a registered medical practitioner with prescribed qualifications and/or experience in a government hospital or in an approved institution.

As a result during the period 1970–85, 222,331 legal abortions were documented. The proportion of legal abortions to live births has also increased progressively from 4·1% in 1970 to 55·3% in 1985 (Table 2).

At the same time acceptance of contraception increased. Family planning in Singapore began in 1949 with the formation of the Singapore Family Planning

Table 2. Legal abortions, Singapore

	Abortions			Abortions	
Year	No.	% of live births	Year	No.	% of live births
1970	1,913	4·1	1978	17,246	43·7
1971	3,407	7·2	1979	16,999	41·7
1972	3,806	7·7	1980	18,219	44·2
1973	5,252	10·9	1981	18,890	44·7
1974	7,175	16·6	1982	19,910	44·8
1975	12,813	32·2	1983	19,000	46·8
1976	15,496	36·2	1984	22,190	53·4
1977	16,443	42·9	1985	23,512	55·3

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Association, a voluntary organization concerned with the health and welfare of mothers and children. Only in 1965 did the Ministry of Health give serious consideration to family planning activities in government clinics and launched its first 5-year plan; three further 5-year plans followed up to 1980 (statistics from Registrar of Births and Deaths, Singapore). The proportion of women in the childbearing years accepting contraception increased from below 60% in 1965 to about 90% to date, and over the decade 1970–80 the proportion of acceptors below 34 years of age increased from 62% to 82% (Singapore Family Planning and Population Board Annual Reports).

Demographic effects

Crude birth rate

The crude birth rate in Singapore was 42·7 per 1000 population in 1957 and subsequently declined sharply. At the end of 1965 it was 29·5 per 1000 population. The introduction of the government-supported National Family Planning programme in 1966 caused the crude birth rate to decline further, to 20 per 1000 in 1970. With the abortion legislation the crude birth rate fell further so that in 1985 it was 16·6 per 1000 population (Fig. 1). This trend occurred in the three major ethnic groups in Singapore.

Infant and maternal mortality rates

The change in childbearing resulting from the introduction of family planning brought a dramatic reduction in both infant and maternal mortality. In Singapore perinatal mortality was 28·3 per 1000 total births in 1957; it fell to 25·5 per 1000 in 1965, then, with the introduction of the National Family Planning programme in 1966 and the abortion legislation in 1970, it decreased sharply to 10·6 in 1985 (Fig. 1).

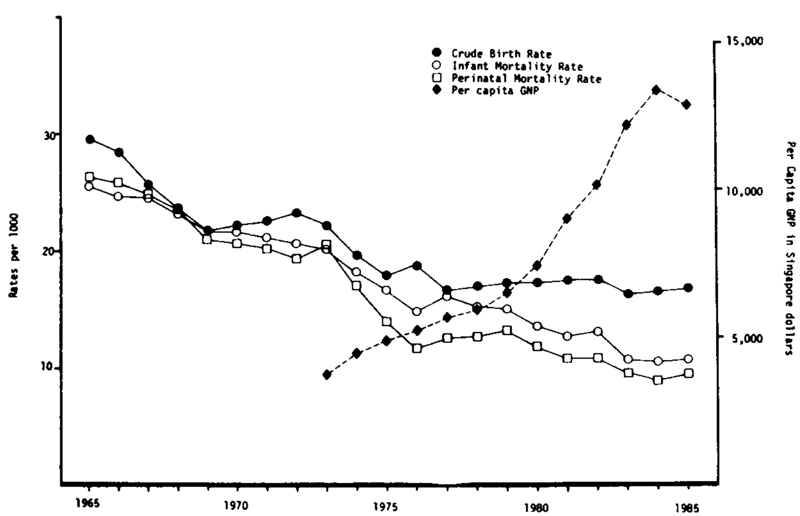

Fig. 1. Crude birth rate, and infant and perinatal mortality rates for Singapore, 1965–85, with change in GNP per head, 1973–85. Source: Singapore Family Planning and Population Board Annual Reports.

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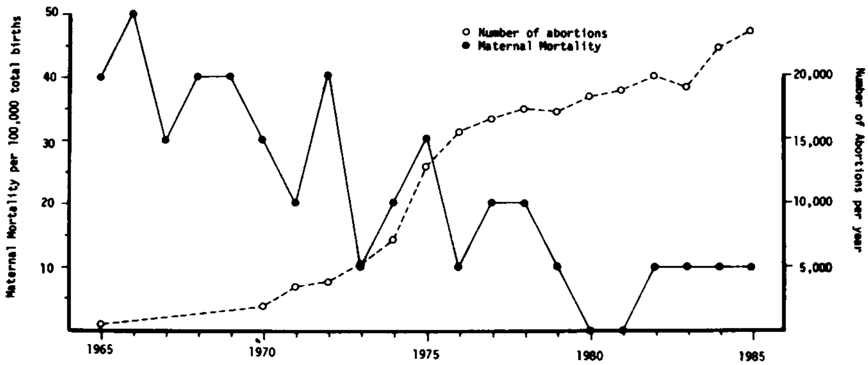

Fig. 2. Trends in maternal mortality rates in Singapore in relation to legal abortion rates. Source: as Fig. 1.

A similar trend also occurred in the infant mortality rate; this was 41·4 per 1000 live births in 1957 and 26·2 in 1965. With the introduction of family planning and legalized abortion, it was further reduced so that in 1985 the infant mortality rate for Singapore was 9·3 per 1000 live births.

Maternal mortality rates similarly declined, from 90 per 100,000 live births in 1957 to 10 per 100,000 live births in 1985 (Fig. 2). Furthermore, with the liberalization of

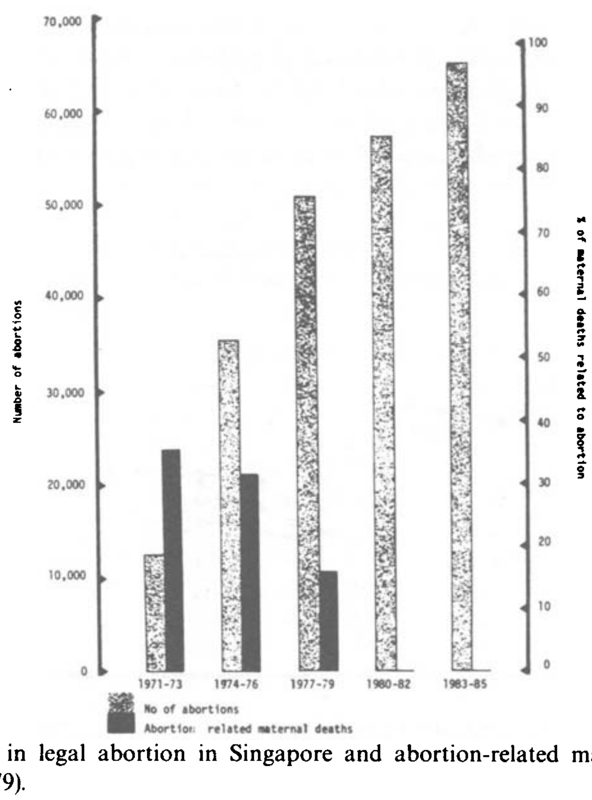

Fig. 3. Trends in legal abortion in Singapore and abortion-related maternal deaths (Lim *et al.*, 1979).

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abortion legislation in Singapore in 1970 and the availability of abortion on demand for pregnancies up to 24 weeks in 1974, there was a progressive fall in the abortion-related deaths from fifteen in 1968–70 to nine in 1974–76, a decline of 40% (Lim *et al.*, 1979), despite the increase in the number of legalized abortions over the same period. There were no abortion-related deaths in the years 1980–85 (Fig. 3).

While it seems that the success of the National Family Planning programme and the legislation on abortion in Singapore has been largely responsible for the decline in maternal, perinatal and infant deaths, at the same time there was an overall improvement in maternal educational status and the socioeconomic circumstances of the family, for the gross national product per head increased from 3849 Singapore dollars in 1973 to 12,865 Singapore dollars in 1985 (Fig. 1). These socioeconomic changes must have contributed to the mortality improvement.

Birth interval

Women who use contraceptives are more successful than other women in spacing their births and preventing unwanted pregnancies. In Singapore the number of women contraceptive acceptors who want more children increased from 67% in 1976 to 88% in 1985. This trend is greatest in women who are of parity 0 or 1, in whom the use of contraception increased from 43% in 1970 to 81% in 1983.

Maternal age of childbearing

In Singapore, among women aged 15–19 years, there was a decline of 59·3% in the births between 1966 and 1985. For women aged over 35 years the decline in the births was 46·1% over the same period (Fig. 4).

In 1985 the prime childbearing age group was 25–29 years and 42·0% of total births were to mothers in this age group; 89·1% of live births were to mothers in the age group 20–34 years. The average age of mothers was 26·8, 27·0 and 28·5 years for Malays, Indians and Chinese respectively in 1985. About 40% of Malay and 34·5% of Indian mothers were in the younger age groups 15–19 and 20–24 years compared with only 19·4% of Chinese mothers. In these two groups the perinatal outcome has remained worse than that in the Chinese population (Viegas, Heng & Ratnam, 1984).

Birth order

In Singapore (Fig. 4) the numbers of fourth and later births fell by almost 90% between 1966 and 1985, and first and second births increased, again an indicator of the success of the National Family Planning programme.

Fertility

Total fertility rate. In Singapore during the period 1957–65 this declined from 6·4 to 4·6 per woman. With the introduction of the government-backed National Family Planning programme in 1966, the decline accelerated (Fig. 5) so that in 1970 the rate was 3·1 per woman and in 1986, 1·6.

Gross reproductive rate. This was 3·186 in 1957. Following the introduction of the government-backed National Family Planning programme in 1966, it declined rapidly to 1·505 in 1970. The decline slowed in 1971, and in 1972 a small rise of 0·5% was recorded; it is likely that those women who earlier had accepted family planning

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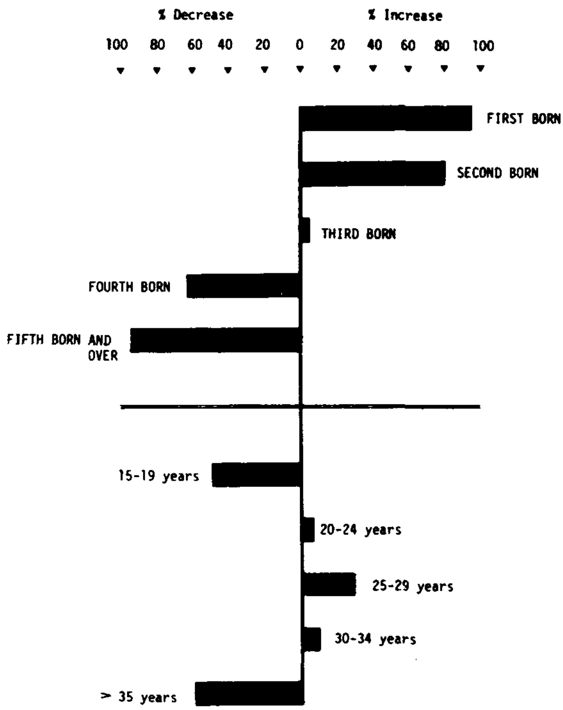

Fig. 4. Changes in fertility patterns in Singapore, 1967–85, by birth order and maternal age. Source: statistics from Registrar of Births and Deaths.

for the purpose of spacing were than having their postponed babies. With the strengthening of the government family planning programmes and the introduction of legalized abortion, voluntary sterilization and disincentive measures in the early 1970s, the rate was brought down to 1·006 in 1975, below the replacement level of 1·025, corresponding to a net reproduction rate of 1·000. The gross reproduction rate continued to decline in the next few years (Fig. 5) and was 0·765 in 1983 (Saw, 1980, 1986).

Thus the decline in fertility from 3·118 in 1957 to below replacement level in 1975 and thereafter is well documented. How far it may be attributed to the effectiveness of national population control programmes is not clear. It certainly began before the initiation in 1966 of the Singapore national programme and can be first detected in 1958, some 10 years after family planning services began to be made available by the private family planning association. But the trends then in progress seem to have been given new impetus by the series of government actions. These, of course, were not operating in isolation, for at the same time certain cultural, social and economic factors making for low reproductive performance were also changing (Chen & Fawcett, 1979; Saw, 1979). Some of these important factors were: (a) rise in family income; (b) increasing labour force participation of women; (c) scarcity of domestic servants, particularly for child care; (d) break-up of the extended family system; (e) rise in educational attainment; (f) changing attitudes towards marriage and

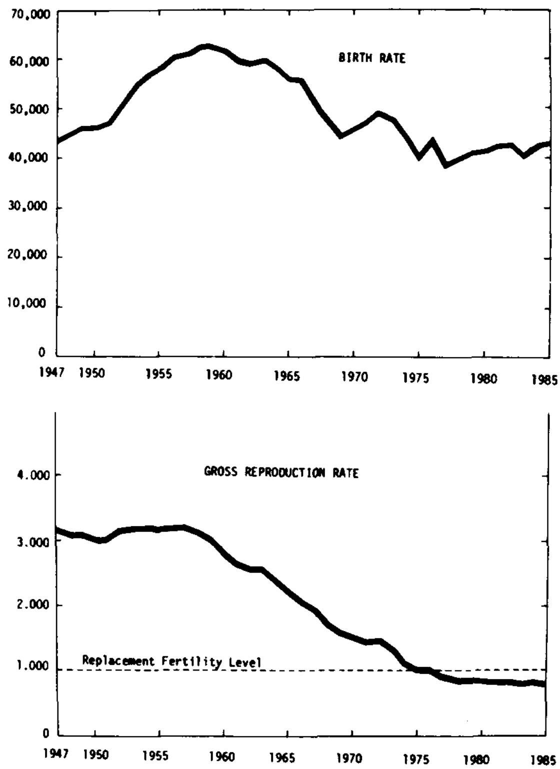

Fig. 5. Birth and gross reproduction rates in Singapore, 1947–85. Source: as Fig. 4.

childbearing, leading to a rise in the age of first marriage from 20 years in 1947 to 25·3 in 1985.

New population policies

The drop in fertility to below replacement level was contributed to mainly by the higher socioeconomic class, more affluent and educated Singaporeans (Saw, 1980). Thus, in 1981, the Singapore government introduced further population measures with the specific aim of regulating the quality of the human population (Saw, 1985).

Graduate mother priority scheme. Graduate mothers were given priority in the scheme of registering pre-primary and primary-one pupils. However, this policy, being so selective, created social dissatisfaction and the government decided to discontinue the scheme in May 1985.

Enhanced child relief scheme. A second pronatalist policy, announced in 1984, designed to encourage better educated women to produce more children was the enhanced income tax relief for the better educated and qualified women. It is very unlikely that this measure will encourage many of the better educated working women to produce more children, because of the very complicated demands they make on the time and effort of the mother both at home and at work. Greater availability of good domestic servants and convenient child care centres may play a more effective role in encouraging better educated working mothers to produce more children.

Sterilization cash incentive scheme. This measure takes the form of a $10,000 cash grant when a mother under 30 years has a tubal ligation after the first or second child. Neither of the parents should have any GCE ordinary level passes and neither of them must be earning more than 750 Singapore dollars.

Government hospital accouchement scheme. The accouchement fees for the four ward classes were increased for the third and subsequent births, the increase being greater for the lower ward classes. The aim is to discourage high parity births among the poorer and less educated segment of the population.

It is difficult to predict at this initial stage whether the new measures will yield the desired results since their effect will take some years to show. From the quantitative point of view, it is likely that on balance the four measures will lead to fewer births rather than more. Thus the new population policies may tend to prolong the fertility trend below the replacement level in the near future.

Ethical considerations

In the implementation of any government policy using incentives or disincentives, ethical issues are inevitable. Two main objectives relating to incentive programmes are: (i) they may infringe on the freedom of choice by influencing the options of parental decision makers, and (ii) they may be unjust.

However, positive incentives can be seen to enlarge options rather than reduce them, and hence serve both freedom and justice. The individual is presented with a choice she did not have in the absence of the incentive. Since she can decline the incentive, she has lost nothing from her previous state but has gained an option—the incentive rather than her normal fertility behaviour.

With negative incentives, the main problem is the possible effect on third party innocents (children) born in disregard of disincentives, for example the disadvantage of second choice schooling. In addition, there is the possibility of psychological or internal effects in the form of intrafamily reactions to innocents on the ground that they cost the family something. Although such effects may not be large, they nevertheless should be safeguarded against.

In general, positive incentives are usually permissible, provided that means of resolving a population problem are not sufficiently effective in their absence and that those who are affected by the incentives possess sufficient knowledge and understand the consequences of accepting them.

Conclusion

Effective family planning, legislation on abortion and voluntary sterilization in Singapore have clearly helped the government to achieve the desired demographic changes within this densely populated city state. Singapore's success in achieving its demographic goals within a legislative framework is unique, though unfortunately Singapore's experience will not apply to developing countries where size, rural conditions, low level of literacy and scarce economic resources prove to be major setbacks in the implementation of such population policies.

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